Customer Shopping Behavior Analysis

A comprehensive analysis of **3,900 transactions** across multiple product categories — uncovering consumer segments, spending trends, and subscription behavior to inform strategic business decisions.

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Dataset at a Glance

3,900

Total Purchases

Rows of transactional data

18

Data Columns

Features per transaction

37

Missing Ratings

Null values in Review Rating

4

Key Feature Groups

Demographics, purchases, habits & ratings

Features span customer demographics, purchase details (item, category, amount, season), and behavioral signals (discounts, promo codes, review ratings, shipping type).

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Data Preparation in Python

01

Data Loading & Inspection

Imported via Pandas; used `describe()` and `df.info()` to verify structure.

02

Missing Data Handling

Imputed missing Review Ratings using the median per product category.

03

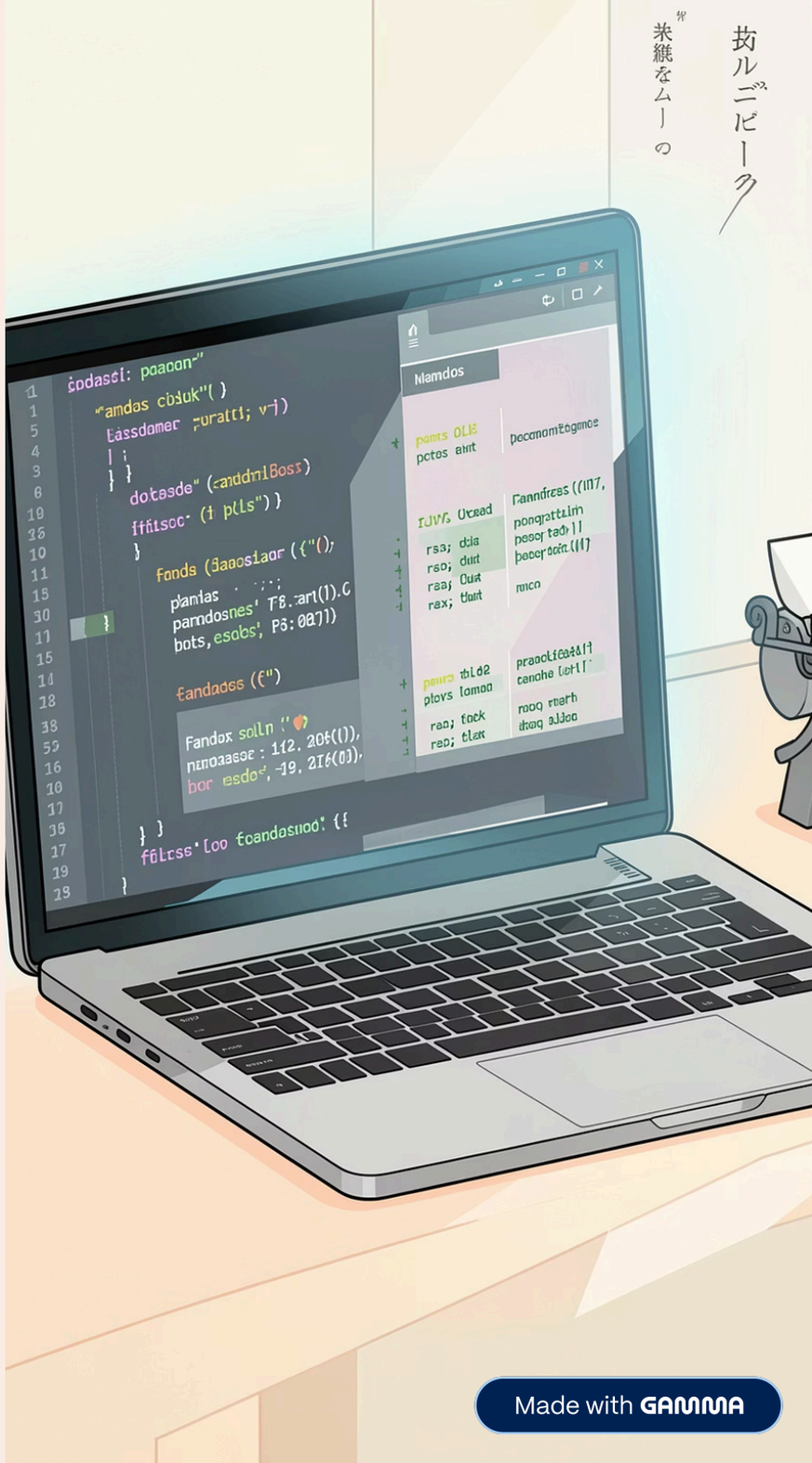
Feature Engineering

Created `age_group` (age bins) and `purchase_frequency_days`; dropped redundant `promo_code_used` column.

04

Database Integration

Loaded cleaned data into PostgreSQL for SQL-based investigation.



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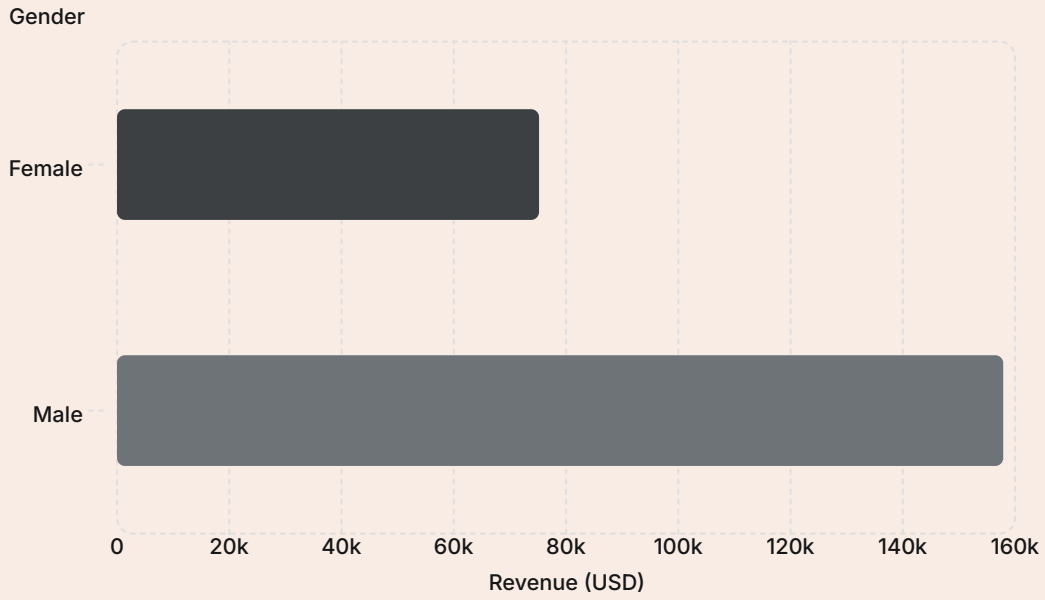
Revenue by Gender



Key Insight

Male customers generated **$157,890** in total revenue — more than double the **$75,191** from female customers, representing a significant spending gap across the dataset.

Male shoppers account for ~68% of total revenue despite representing a similar transaction count.

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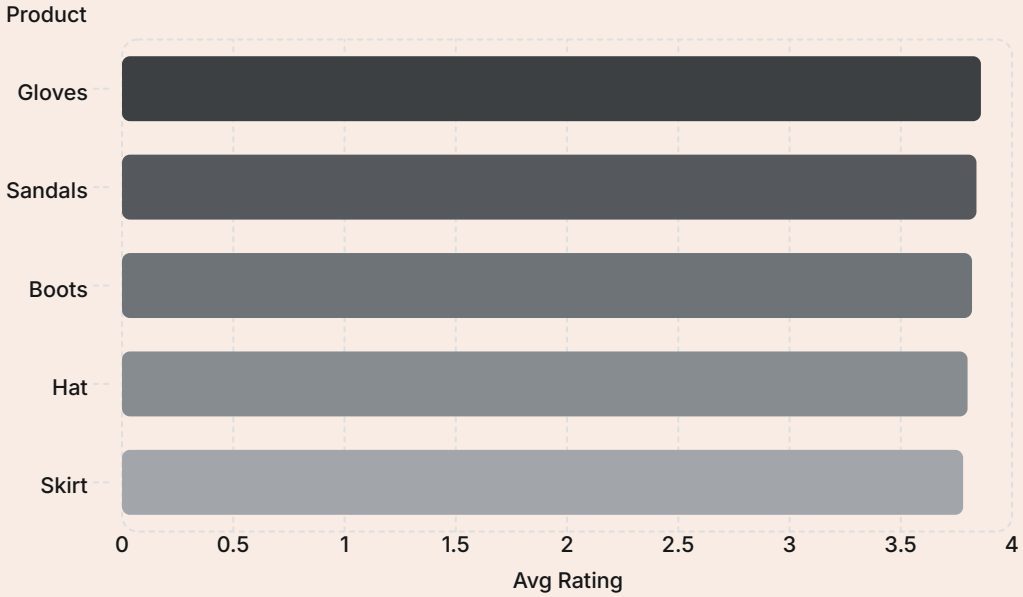
Product Ratings & Discount Patterns

Top 5 Products by Average Rating



Most Discounted Products

These five items had the highest discount rates. Notably, Hat and Sneakers also rank among the most popular products in their categories.

Hat — 50%

Sneakers — 49%

Coat — 49%

Sweater — 48%

Pants — 47%

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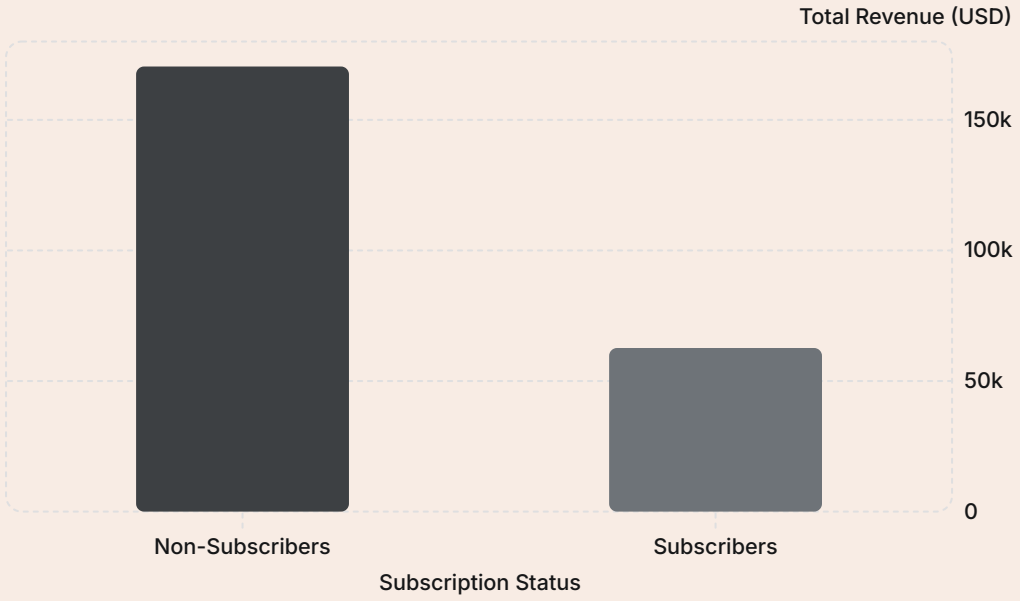
Subscribers vs. Non-Subscribers



Key Findings

- **73% non-subscribers** (2,847 customers) vs. 27% subscribers (1,053)
- Average spend is nearly identical: **$59.87** vs. **$59.49**
- Non-subscribers drive **$170K** in revenue; subscribers contribute **$62.6K**

Subscribers spend similarly per transaction but are far fewer — a clear growth opportunity.

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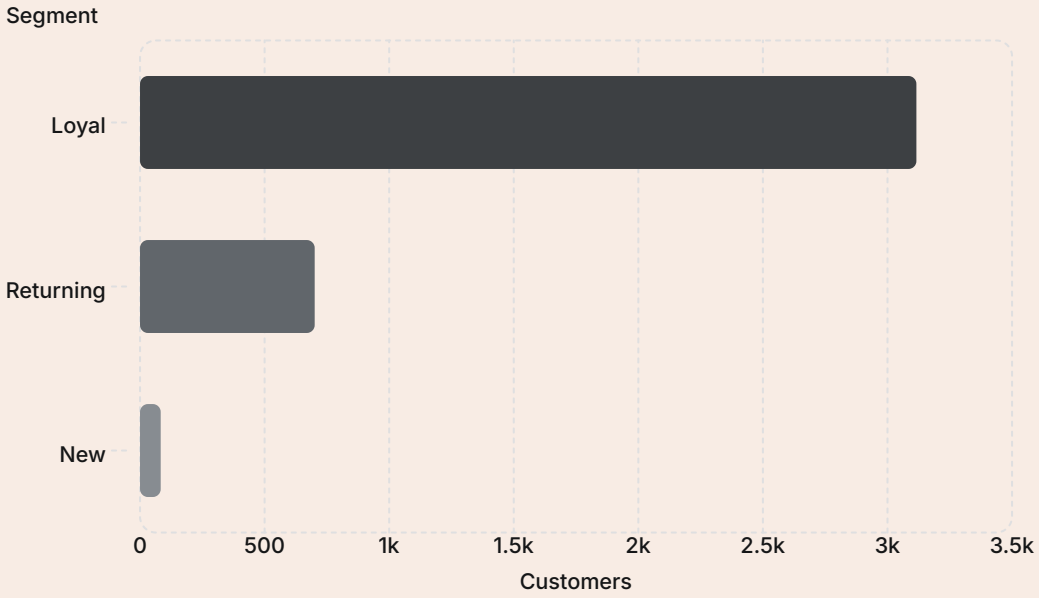
Customer Segmentation



Segment Breakdown

Of customers with more than 5 prior purchases, **958 are subscribers** vs. **2,518 non-subscribers** — repeat buyers show strong subscription potential.

Loyal — 3,116

80% of base; high repeat purchase history

Returning — 701

18%; potential to convert to Loyal

New — 83

2%; onboarding opportunity

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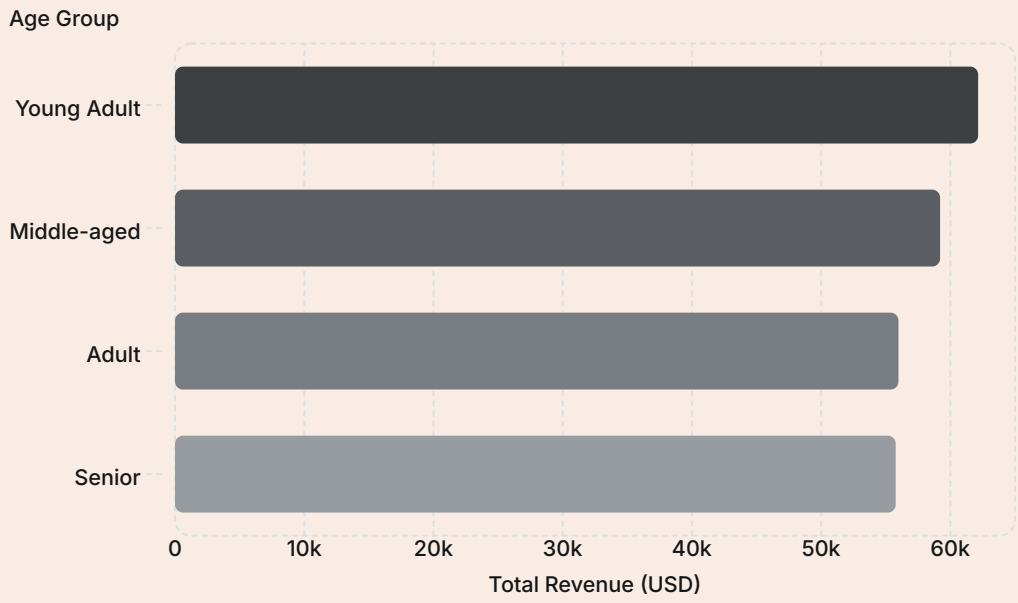
Revenue by Age Group



Age Group Insights

Young Adults (18-30) lead revenue at **$62,143**, with a gradual decline across older groups. Revenue distribution is relatively even — suggesting broad product appeal across all demographics.

- Young Adult: **$62,143**
- Middle-aged: **$59,197**
- Adult: **$55,978**
- Senior: **$55,763**

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Power BI Dashboard

Dashboard KPIs Visualizations

- Revenue & Sales by Category (Clothing, Accessories, Footwear, Outerwear)
- Revenue & Sales by Age Group
- % Customers by Subscription Status (27% Yes / 73% No)

Interactive filters: Subscription Status, Gender, Category, Shipping Type.

3.9K

Total Customers

$59.76

Avg Purchase Amount

3.75

Avg Review Rating

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Business Recommendations

Increase Subscriptions

Highlight exclusive subscriber benefits to convert the 73% non-subscriber base.

Loyalty Programs

Incentivize Returning customers (701) to reach Loyal status with targeted rewards.

Review Discount Policy

Balance margin control with sales growth — especially on high-discount items like Hats (50%) and Sneakers (49%).

Targeted Marketing

Focus campaigns on express-shipping customers and high-revenue age groups (Young Adults, Middle-aged).

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